UNIVERSITY OF LONDON

MSc EXAMINATION 2023

For Internal Students of
Royal Holloway

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CS5960: ARTIFICIAL INTELLIGENCE PRINCIPLES AND TECHNIQUES – for FIRST-SIT CANDIDATES
CS5960R and CS5960JR: ARTIFICIAL INTELLIGENCE PRINCIPLES AND TECHNIQUES – for RESIT CANDIDATES

Time Allowed: **TWO hours**

Please answer **ALL** questions

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1. (a) Consider the graph in Figure 1. The initial state is 1. Run the depth-first search algorithm on this problem. Draw the search graph and annotate the nodes with the order in which states are selected for expansion.

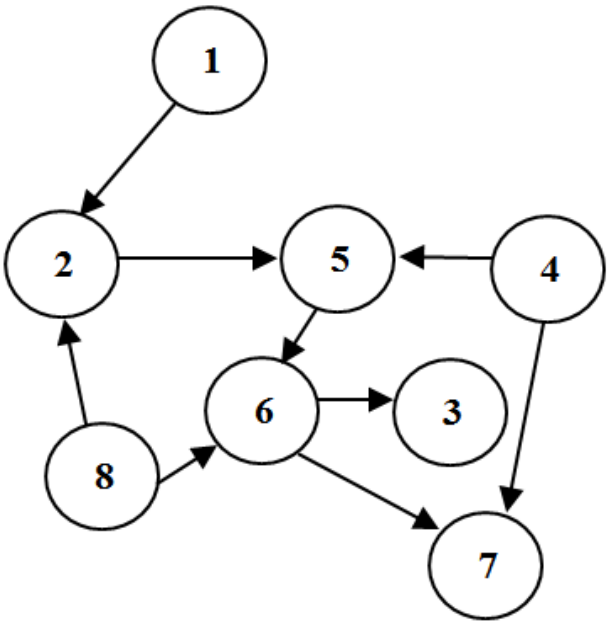


Figure 1: Graph for depth-first search

[6 marks]

(b) Consider the graph in Figure 2. Show the order of nodes visited for breadth-first search and depth-first search.

[6 marks]

(c) What are the time and space complexities of the depth-first search and iterative deepening search algorithms? [5 marks]

(d) Consider the graph in Figure 3. The initial state is A (depth=0). Show the order of nodes visited using the iterative deepening search from depth=0 to depth=3. (You need to list the order of nodes visited for each depth.)

[8 marks]

(e) Consider the missionaries and cannibals problem, starting with two missionaries, two cannibals, and one boat on the right bank. The boat can carry at most two people on each trip. The task is to transport all the missionaries and cannibals to the left bank. If there are both missionaries and cannibals

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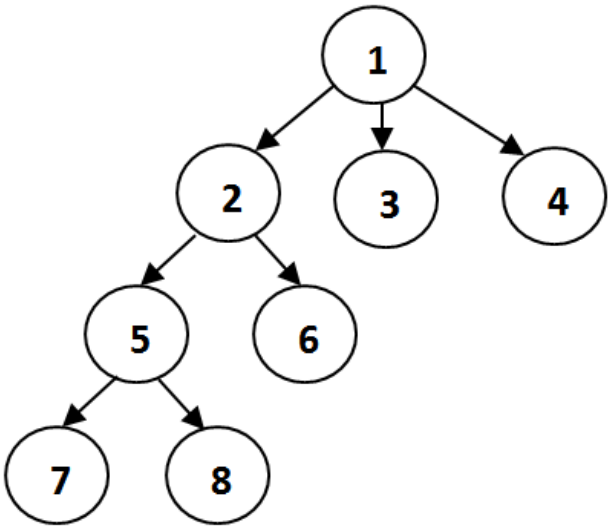


Figure 2: Graph

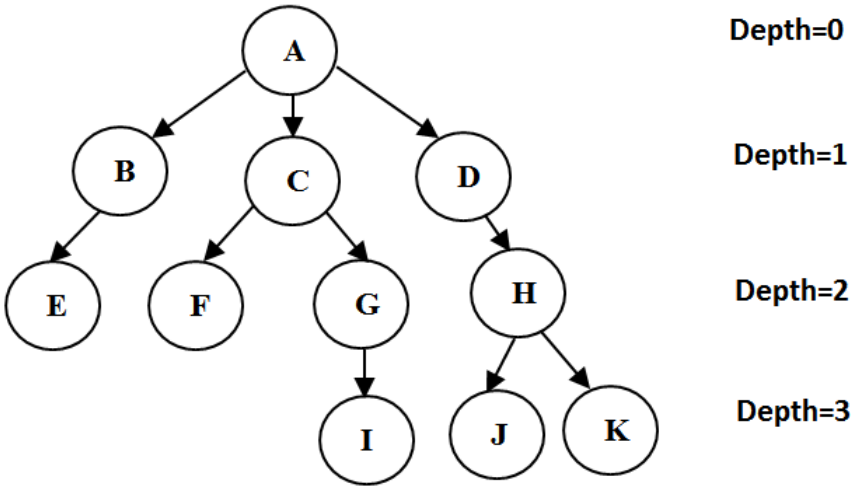


Figure 3: Graph for iterative deepening search

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on a bank and the missionaries are outnumbered by the cannibals, the task is failed and the game terminates.

Formulate the problem as a *state space* problem and use the *breadth-first search* algorithm to find a solution for the problem. Draw the search graph and annotate each node with the order in which states are selected for expansion. Each state should be represented as a tuple $s = (m, c, b)$, such that m, c and b stand for the numbers of missionaries, cannibals and boat on the right bank, respectively. You do not need to expand the duplicate nodes. If the choice of the next state to be expanded is not unique, follow the rules below to break ties:

i. Expand the state with smaller m value first. For example, suppose two states $s_1 = (m_1, c_1, b_1)$ and $s_2 = (m_2, c_2, b_2)$ are in the frontier; if $m_1 < m_2$, expand s_1 before s_2.

ii. If m_1 is equal to m_2, expand the state with smaller c value first. For example, if $m_1 = m_2$ and $c_2 < c_1$, expand s_2 first.

[15 marks]

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2. (a) Explain the relationship between the A* algorithm and the Uniform Cost Search algorithm, and the relationship between the A* algorithm and Greedy Best-First Search algorithm. [8 marks]

   (b) Consider the 8-puzzle problem, given the initial state and the goal state shown in Figures 4 and 5, respectively.

2	8	3
1	6	4
7		5

Figure 4: Initial state for 8-puzzle

1	2	3
8		4
7	6	5

Figure 5: Goal state for 8-puzzle

Use the A* algorithm to solve this 8-puzzle problem, with the following evaluation function $f(n)$.

$f(n) = g(n) + h(n)$

where,

$g(n)$ = Steps taken from the start node to get to the current node (i.e. the cost of the search so far)

$h(n)$ = Number of misplaced tiles.

Generate the search tree. For each node, provide cost details for $g(n)$, $h(n)$ and $f(n)$, respectively. [28 marks]

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3. Consider the following constraint network: $\gamma = (V, D, C)$:

   - Variables: $V = \{a, b, c, d, e, f, g\}$
   - Domains: for all $u \in V, D_u = \{1, 2, 3, 4, 5, 6\}$
   - Constraints: $a = 2d,\ g = 3d,\ d = b,\ e = a - 3,\ c = b + 3,\ f = e + 2$

   Run the AC-3 algorithm on the given constraint network. For each iteration of the while-loop of the algorithm, give the content of the data structure M at the start of the iteration, the pair (u, v) removed from M, the domain D_u of u after the call to $Revise(\gamma, u, v)$, and the pairs (w, u) added into M. The $Revise()$ function enforces consistency of one variable relative to its children.
   Note: Initialize M as a lexicographically ordered list (i.e., (a, b) would be before (a, c), both before (b, a) etc., if any of those exist). After initialization, use M as a FIFO (first-in-first-out) queue, i.e., always remove the first (oldest) element from the queue and add new elements to the end of the queue.

   [24 marks]

END

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